EXPERIMENT - 2

Aim : studys of different types of network cables

i) UNSHIELDED TWISTED PAIR (UTP) CABLE:

* Category 3:
  max Data Transmission : 10bps
  Application : 10 Base - T ethernet

* Category 5:
  max Data transmission : upto 100 Mbps
  application : Fast Ethernet, giga ethernet

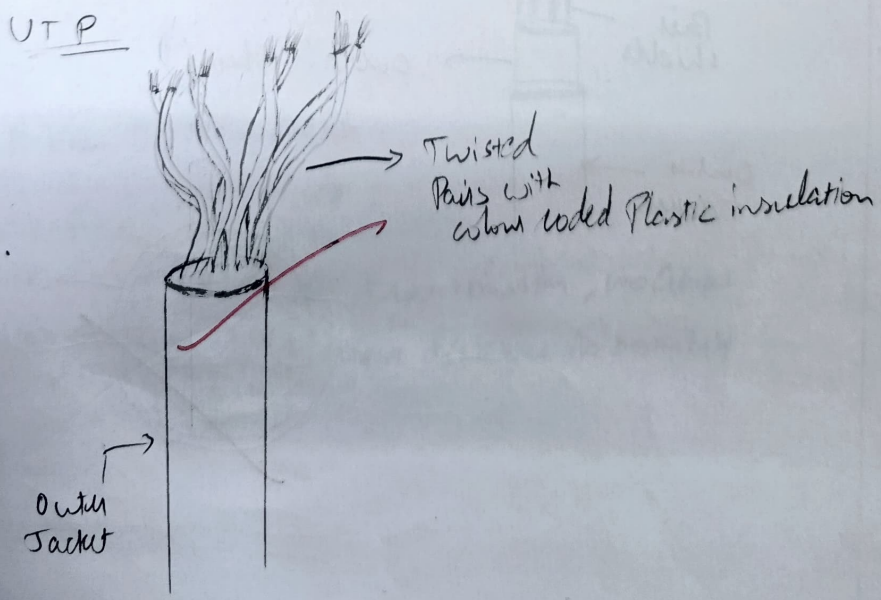
* category 5e:
  application : fast ethernet, gigabit ethernet
  max Data Transmission : 1 Gbps

Advantage : * Cheaper
* easier installation

Disadvantage :
* More Prone to EMI

Diagram:

UTP



ii) Sheilded Twisted Pair (STP) cable:

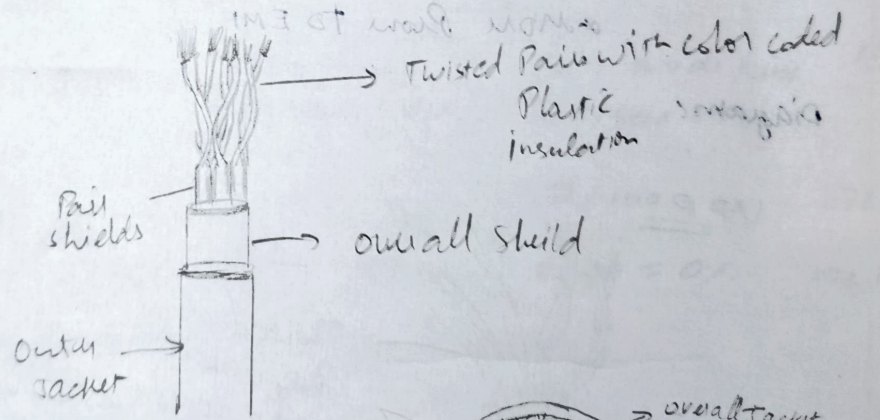
* Catagory 6/6a:
  - Max Data Transmission: 10 Gbps
  - Application: 10G Ethernet

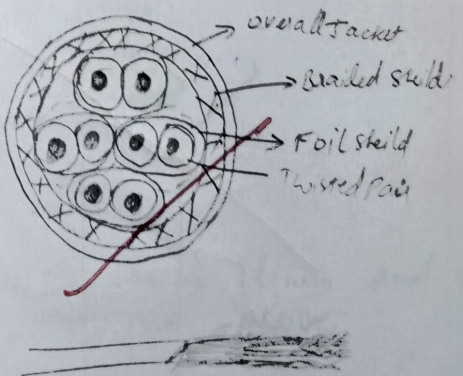
* Category 7:
  - Max Data Transmission: 40 Gbps
  - Application: 10G Ethernet (10Gbase)

Advantages:
* Sheilded
* Faster than UTP

Disadvantages
* Expensive
* Installation Difficult.

Diagram





iii) Coaxial Cable:

* RG-6
  RG-59 :
  RG-11

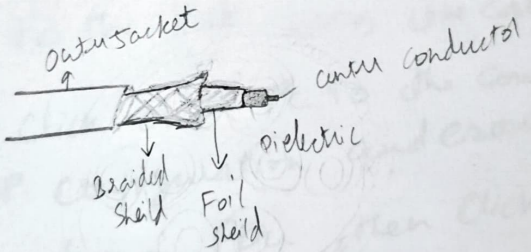
Max Data Transmission: 10 - 100 mbps

Application: speed of signal 500m

Advantages:
* High Bandwidth
* Versatile
* Immune to interface

Disadvantages:
* Limited distance
* Bulky
* cost high

Diagram



iv) Fiber Optics Cable:

* single mode:
multi mode:
max para transmission, 100 Gbps
Application: max distance is 100 km

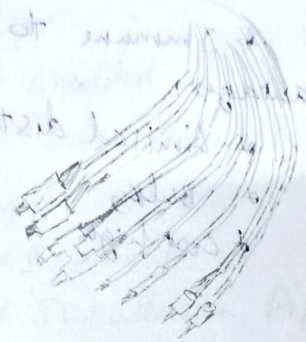
Advantage:

* High Speed
* High security

Disadvantage:

* Expensive
* Required skills

Diagram



Result: Successfully analysed the study of different types of network cables